GOAL

HEART DISEASE PREDICTION



PNEUMONIA DETECTION FROM X-RAY



PERFORM HEALTH MONITORING OPERATIONS







REQUIREMENT

FUNCTIONAL

- The system shall have a login and sign-up feature
- Model that can predict heart disease
- Model that can perform Pneumonia Detection on an X-ray image

REQUIREMENT

FUNCTIONAL

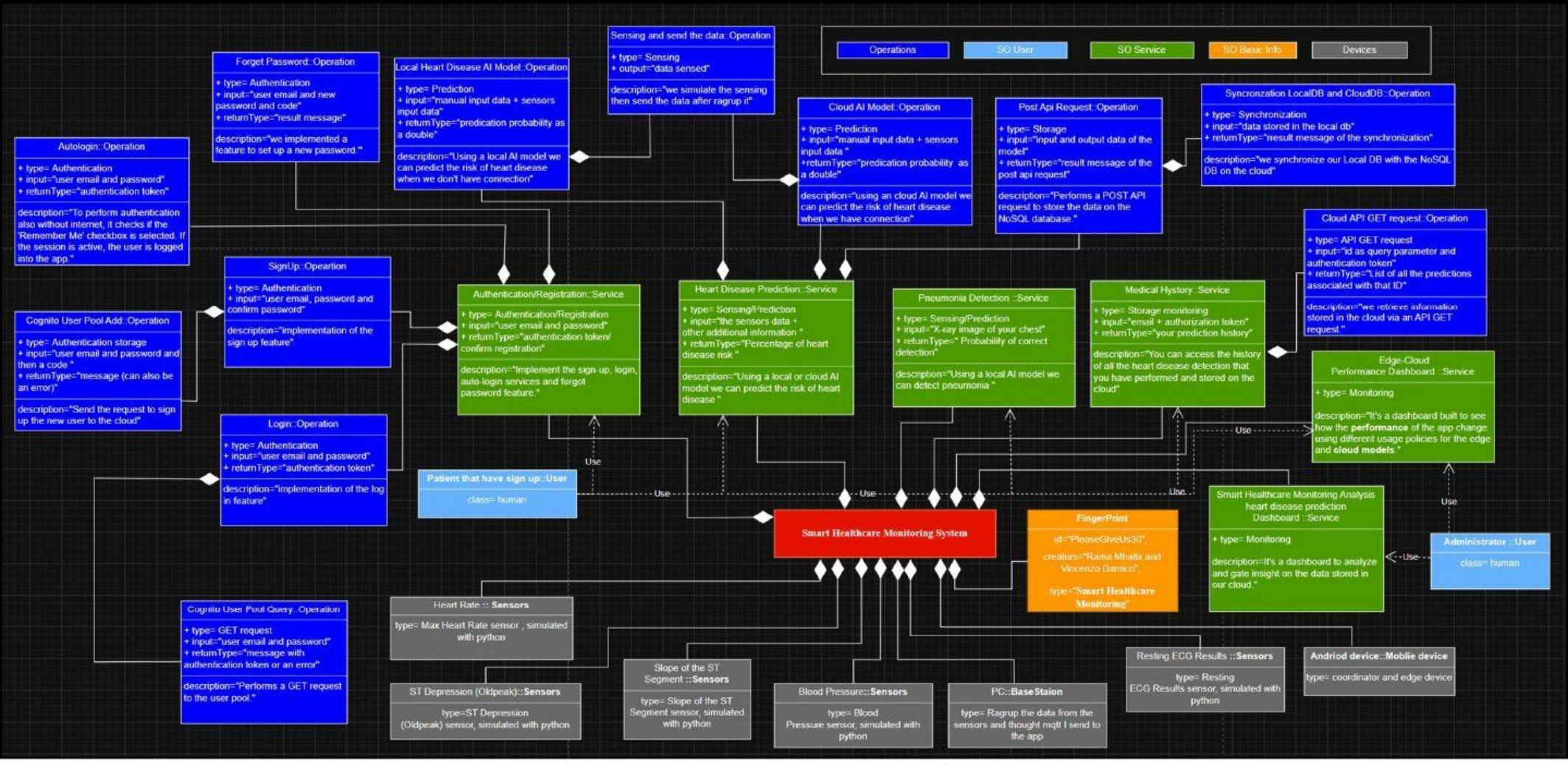
- Store the medical history related to the heart disease predictions
- Work without internet connection
- Analyze and retrieve insight about our system data and performance

REQUIREMENT

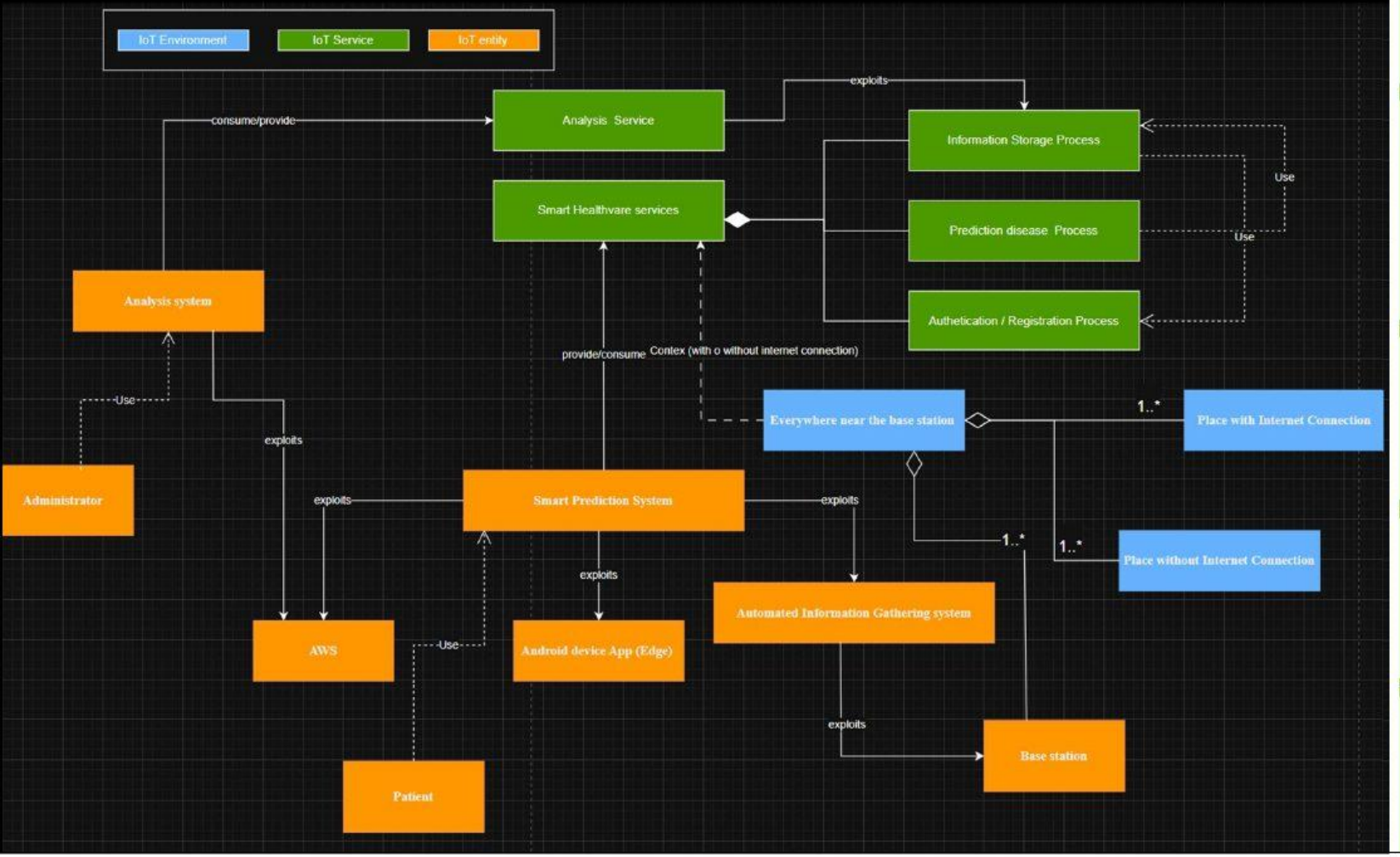
NON-FUNCTIONAL

- Manual and automatic mechanism to insert data
- Auto-login shall be performed even without an internet connection
- Heart disease model shall run on cloud and locally
- The app has to be fast and reliable

ACOSO-METH ANALYSIS MODEL



ACOSO-METH DOMAIN MODEL



Analysis Service

Services output	Data Insight Dashboard Performance Insight DashBoard
Services Input	Prediction records (Input + Output) / Simulations result

Smart Healthcare Services

Services output	Heart disease or X-ray prediction store and associate them with a user
Services Input	Sensors data or X-ray chest image + user credentials

Information Storage Process

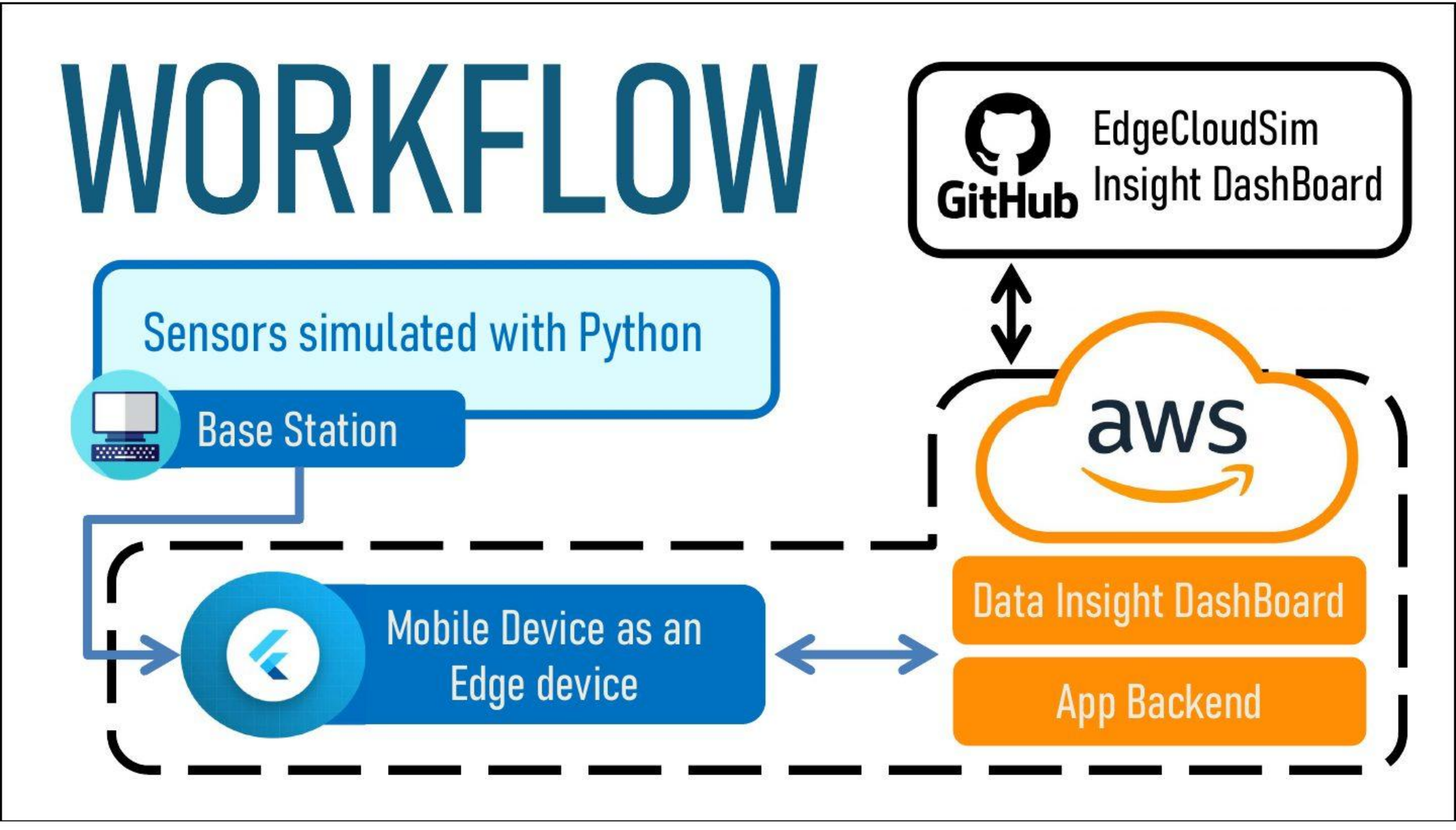
Process Description	Store the heart disease detection that the user have performed and stored

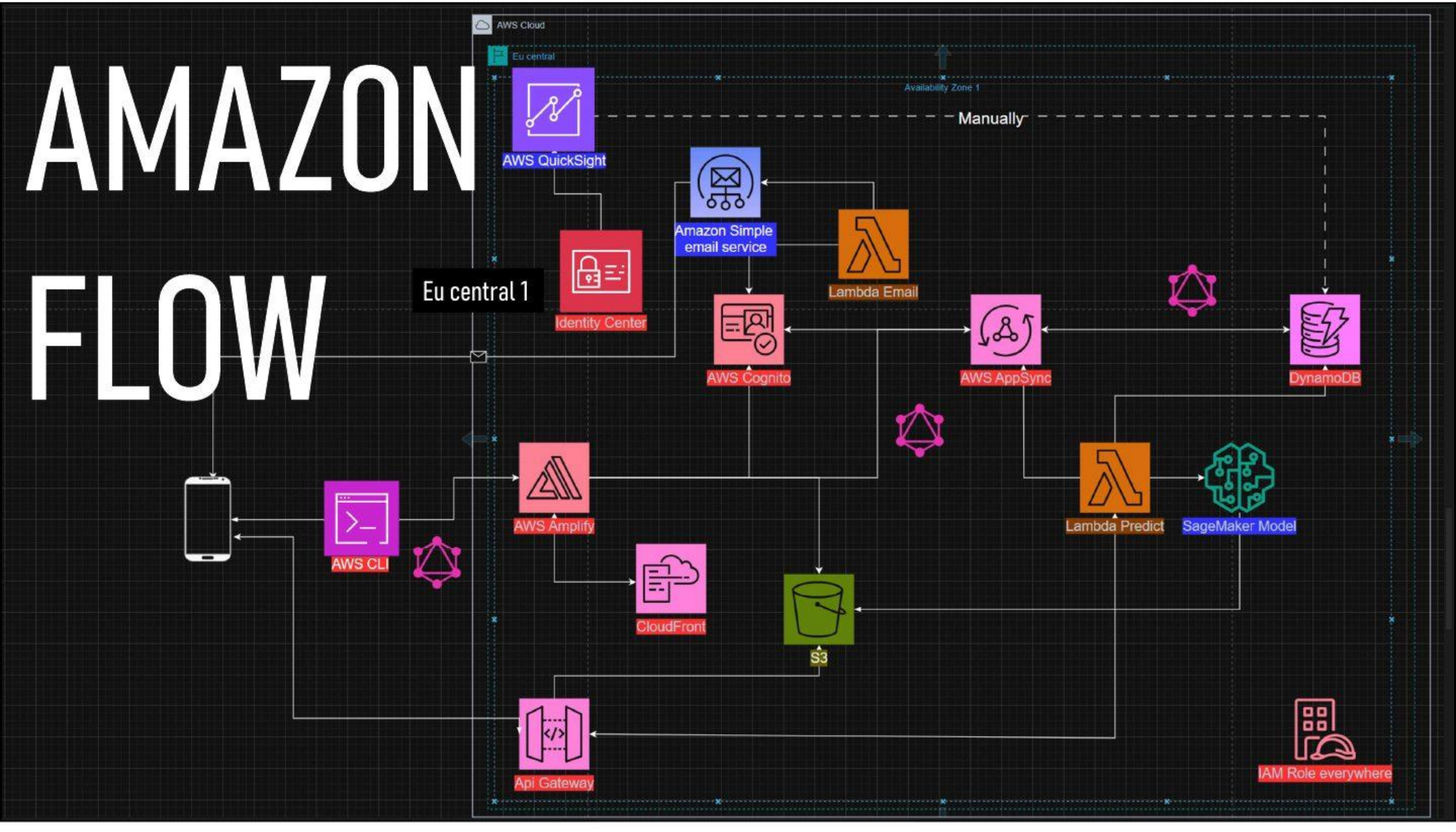
Authetication Registration Process

Process Description	Implement the sign-up, login, auto-login services and forgot password feature

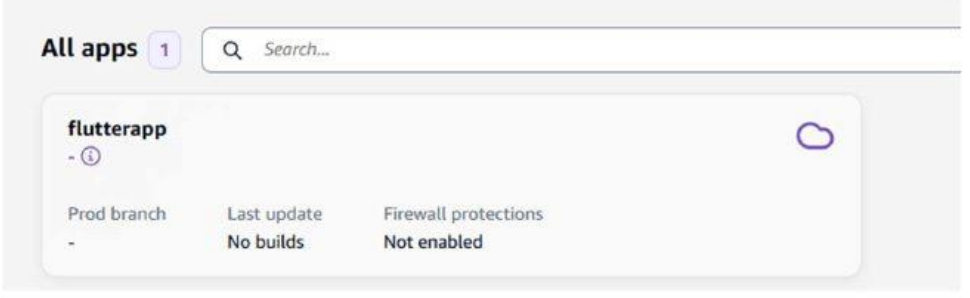
Prediction Disease Process

Process Description	The process can perform heart disease prediction and X-ray prediction


WORKFLOW
GitHub
EdgeCloudSim
Insight DashBoard
Sensors simulated with Python
Base Station
aws
Mobile Device as an
Edge device
Data Insight DashBoard
App Backend


AMAZON
FLOW
AWS Cloud
Eu central
Availability Zone 1
Manually
AWS QuickSight
Amazon Simple email service
Lambda Email
Eu central 1
Identity Center
AWS Cognito
AWS AppSync
DynamoDB
AWS CLI
AWS Amplify
Lambda Predict
SageMaker Model
CloudFront
S3
Api Gateway
IAM Role everywhere

AWS Amplify



Dynamo AppSync Cognito S3

Configurated

AWS CLI IAM User

Amazon SageMaker AI



Notebook

Model

Train

Linked to

Lambda

API Gateway

```python
import json
import boto3

# Create a SageMaker runtime client
runtime = boto3.client('sagemaker-runtime')

def lambda_handler(event, context):
    try:
        # ✅ Handle both test and actual request formats
        if 'body' in event:
            body = json.loads(event['body'])
        else:
            body = event  # For test events sent directly

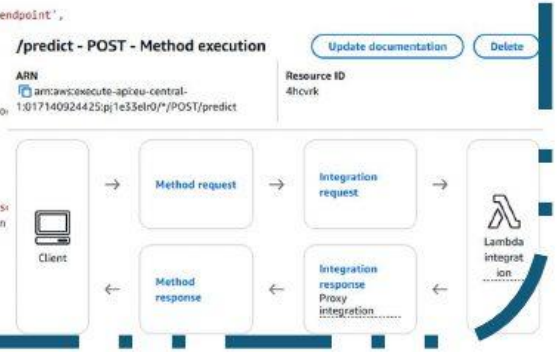
        # ✅ Ensure 'features' exists
        if 'features' not in body:
            return {
                'statusCode': 400,
                'body': json.dumps({'error': "Missing 'features' in request"})
            }
        features = body['features']
        # ✅ Convert to SageMaker payload format
        payload = json.dumps({"instances": [features]})

        # ✅ Call SageMaker endpoint
        response = runtime.invoke_endpoint(
            EndpointName='heart-disease-model-ai-cloud-endpoint',
            ContentType='application/json',
            Body=payload
        )

        # ✅ Parse prediction result
        result = json.loads(response['Body'].read().deco
        prediction = result['predictions'][0][0]

        # ✅ Return success response
        return {
            'statusCode': 200,
            'headers': {'Content-Type': 'application/js
            'body': json.dumps({'prediction': prediction
        }

    except Exception as e:
        return {
            'statusCode': 500,
            'body': json.dumps({'error': str(e)})
        }
```

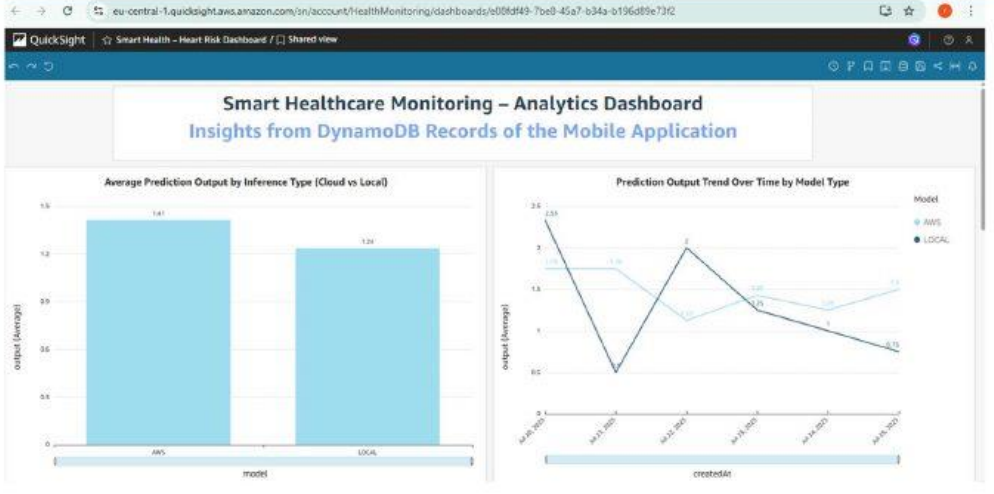


Avg. latency = 640ms

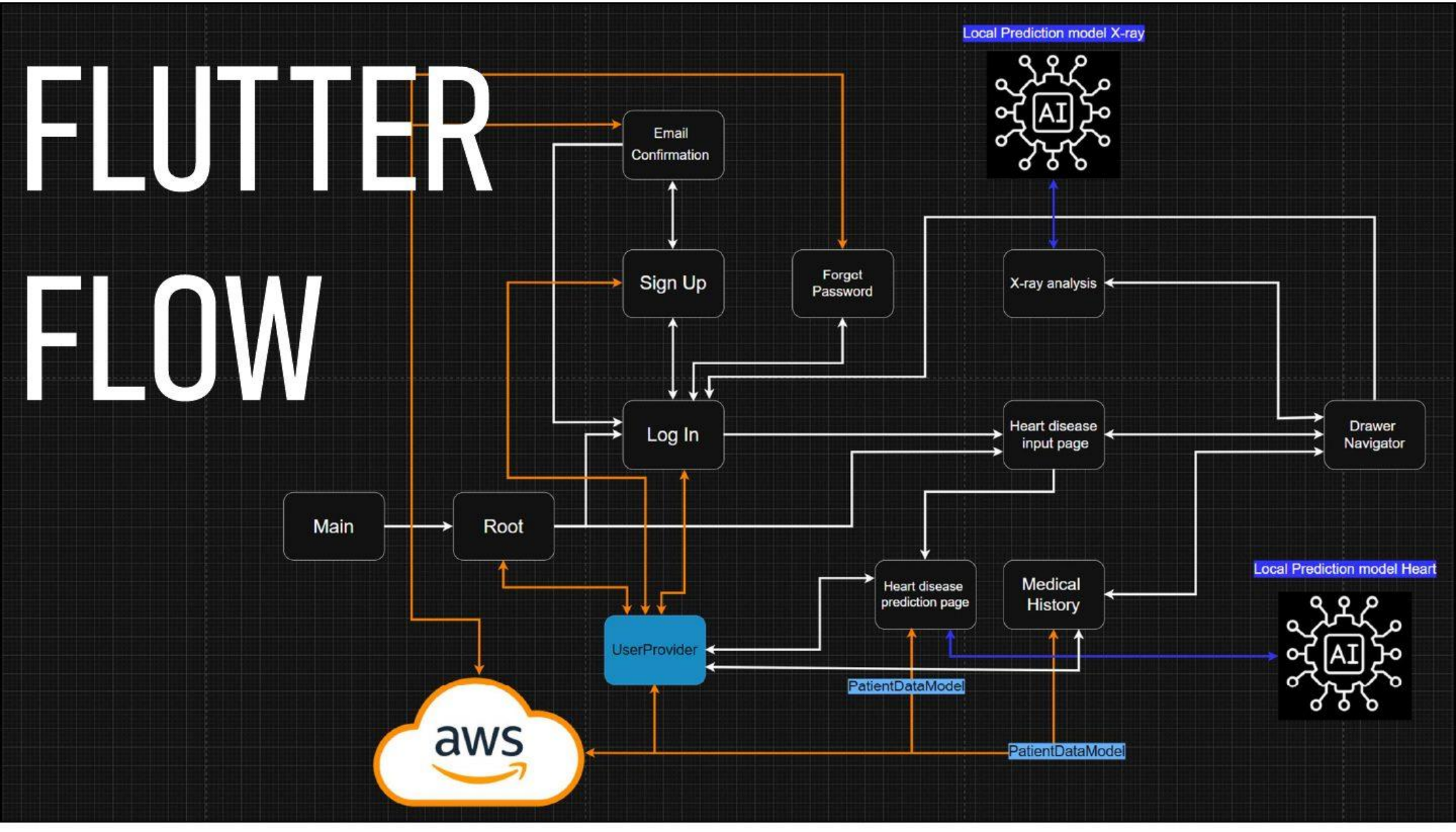
Accuracy = 93.21%

EdgeCloudSim Dashboard

	AI_BASED	RANDOM	PREDICTIVE	GAME_THEORY
Processing Time	Better (slightly lower)	Slightly better	Slightly better	Slightly better
Network Delay	Better (always lower)	A bit better	Slight edge	Slightly better
Failed Tasks (%)	More stable	Better at some loads	Clearly better	Slightly better at times
Service Time	Consistently lower	Slightly better	Slightly better	Consistently better



QuickSight Dashboard


FLUTTER
FLOW
Local Prediction model X-ray
AI
Email Confirmation
Sign Up
Forgot Password
X-ray analysis
Log In
Heart disease input page
Drawer Navigator
Main
Root
Heart disease prediction page
Medical History
Local Prediction model Heart
AI
UserProvider
PatientDataModel
aws
PatientDataModel

kaggle

LOCAL AI MODELS

Avg. latency = 19 ms

Accuracy = 85.25%





MQTT

